

An attained absolute determinant maximum

MathlibAnnex.DeterminantFrame.exists_maximizingFrame

theorem

Selects a maximizing dual frame for a specified basis.

Statement

Let $n \in \mathbb{N}$, let E be a finite-dimensional real normed vector space, and fix an ordered basis $b = (b_j)_{j=1}^n$. Write $C_b : E \rightarrow \mathbb{R}^n$ for the coordinate isomorphism, defined by $x = \sum_{j=1}^n (C_b x)_j b_j$, and give its coordinate space the sup norm $\|c\|_\infty = \max_j |c_j|$ (zero in dimension zero). A frame is an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals, with no independence assumed. Its matrix and evaluation map are

$$M_F = (\ell_i(b_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F C_b x.$$

The admissible frames form the product of closed dual unit balls

$$\mathcal{U} = \{F : \|\ell_i\|_{E^*} \leq 1 \text{ for every } i\}.$$

The norm on E need not be a coordinate sup norm. There is a frame $F_* \in \mathcal{U}$ such that

$$\forall F \in \mathcal{U}, \quad |\det M_F| \leq |\det M_{F_*}|.$$

Assumptions

The space is finite-dimensional over $\mathbb{R}$ and b is its specified basis indexed by n elements. Each admissible row lies in the closed operator unit ball. No independence of the varying frames, positive dimension, or uniqueness of a maximizing frame is assumed.

Conclusion

One admissible frame attains the maximum of the absolute determinant. The numerical maximum $D_b = |\det M_{F_*}|$ depends on b , although its value does not depend on which maximizer is chosen. Positivity is a separate result.

Proof route

Use compactness of the finite product of dual balls and continuity of the absolute determinant.

▼ Proof steps

1. **Supply the compact domain.** The dual space E^* is finite-dimensional, hence its closed unit ball $\overline{B}_{E^*}(0, 1)$ is compact. The compact dual unit ball and finite product of these balls give

$$\mathcal{U} = \prod_{i=1}^n \overline{B_{E^*}}(0, 1) \quad \text{compact.}$$

The zero-frame witness is $\ell_i = 0$ for every i , so the domain is nonempty, also when $n = 0$.

2. **Maximize the correct function.** Every entry $F \mapsto \ell_i(b_j)$ is continuous, and the determinant is a finite sum of products of entries. Therefore $F \mapsto |\det M_F|$ is continuous, as in continuity of the frame determinant. Apply the extreme-value theorem to this continuous real function on the nonempty compact set $\mathcal{U}$. Its output is $F_* \in \mathcal{U}$ with the displayed comparison for every $F \in \mathcal{U}$. This is an absolute maximum; no sign of $\det M_{F_*}$ is asserted.

The selected maximizer is named by the choice of maximizing frame and its value by the attained maximum. The coordinate identity $T_F = M_F C_b$ is evaluation in basis coordinates. These names do not add uniqueness or a nonzero determinant to this theorem.

Main citations

- compact dual unit ball — `MathlibAnnex.DeterminantFrame.isCompact_unitRowSet`
- finite product of these balls — `MathlibAnnex.DeterminantFrame.isCompact_unitFrameSet`
- zero-frame witness — `MathlibAnnex.DeterminantFrame.unitFrameSet_nonempty`
- continuity of the frame determinant — `MathlibAnnex.DeterminantFrame.continuous_frameDeterminant`
- the choice of maximizing frame — `MathlibAnnex.DeterminantFrame.maximizingFrame`
- the attained maximum — `MathlibAnnex.DeterminantFrame.determinantMaximum`
- evaluation in basis coordinates — `MathlibAnnex.DeterminantFrame.frameCoordinates_eq_mulVec`
- Exact declaration and proof — `MathlibAnnex.DeterminantFrame.exists_maximizingFrame`
- Ordered families of continuous linear functionals — `MathlibAnnex.DeterminantFrame.Frame`
- Coordinates in the chosen basis — `MathlibAnnex.DeterminantFrame.coordinateEquiv`
- Inverse coordinate map — `MathlibAnnex.DeterminantFrame.inverseCoordinateMap`
- Matrix of a dual frame — `MathlibAnnex.DeterminantFrame.frameMatrix`
- Determinant in the chosen basis — `MathlibAnnex.DeterminantFrame.frameDeterminant`
- Closed dual unit ball — `MathlibAnnex.DeterminantFrame.unitRowSet`
- Admissible dual frames — `MathlibAnnex.DeterminantFrame.unitFrameSet`

Lean source signature (exact)

```
/-- Existence of an attained absolute determinant maximum. -/
theorem exists_maximizingFrame (b : Basis (Fin n) ℝ E) :
  ∃ B ∈ unitFrameSet n E,
    ∀ C ∈ unitFrameSet n E,
      |frameDeterminant b C| ≤ |frameDeterminant b B|
```

The basis is the explicit input. The result after the colon is the existence and maximizing property of one frame.

In the source	Mathematical meaning
<code>n; b : Basis (Fin n) ℝ E</code>	A nonnegative integer n and the ordered basis $(b_j)_{j=1}^n$ of E .

In the source	Mathematical meaning
<code>[NormedAddCommGroup E] [NormedSpace ℝ E] [FiniteDimensional ℝ E]</code>	The ambient E is a finite-dimensional real normed vector space, with its given norm (not necessarily a coordinate sup norm). These hypotheses occur in the separate surrounding excerpt.
<code>Fin n → ℝ</code>	The coordinate space $\mathbb{R}^n$ with the sup norm. Lean indexes the coordinates by $0, \dots, n-1$; the formulas here use $1, \dots, n$.
<code>Frame n E</code>	The ordered families $F = (\ell_i)_{i=1}^n$ with $\ell_i : E \rightarrow \mathbb{R}$ continuous and real linear. No independence is part of this type.
<code>unitFrameSet n E</code>	The set $\mathcal{U} = \{F : \ \ell_i\ _{E^*} \leq 1 \text{ for every } i\}$. Its elements are families of functionals, not points of E .
<code>∃ B ∈ unitFrameSet n E</code>	There exists a frame $F_* \in \mathcal{U}$. This chosen maximizing frame is called B in the source.
<code>∀ C ∈ unitFrameSet n E</code>	The comparison is required for every $F \in \mathcal{U}$, called C in the source.

In the source	Mathematical meaning
<code>frameDeterminant b C; frameDeterminant b B</code>	Respectively $\det M_F$ and $\det M_{F_*}$, with $(M_F)_{ij} = \ell_i(b_j)$.
<code> frameDeterminant b C ≤ frameDeterminant b B </code>	The output comparison $ \det M_F \leq \det M_{F_*} $. The bars are absolute values, so no orientation sign is fixed.

Here $\forall$ means “for every”, $\exists$ means “there exists”, and $\in$ means set membership. Juxtaposition such as `unitFrameSet n E` is function application, not multiplication.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 47–49:

```
variable {n : ℕ}
variable {E : Type u} [NormedAddCommGroup E] [NormedSpace ℝ E]
[FiniteDimensional ℝ E]
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 37–49. The full original context is retained with the source evidence.

▼ **Exact content identity**

Declaration: `MathlibAnnex.DeterminantFrame.exists_maximizingFrame`

Accepted content SHA-256: `dbb5325a3743a839031e5cede9e311bc25abe486e04b062d2f3d4c544721760b`

Accepted source guide SHA-256: `2ba8385db97c730d988234e7310a00804bf016106832f42d777a594bd7e1996c`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`